WATERLOO ROCKETRY

Eridium Liquid Rocket Engine
Static Fire 2

Static Fire Operations Procedure

Compiled on 2024-05-18

Static Fire Test Operations Procedure

Contents

This document contains the following four procedures:

- The *Fill System Leak Check Procedure* comprises steps for verifying the integrity of the system plumbing and proper operation of the data acquisition and test control systems, using nitrogen gas.
- The *Fill System Check Abort* comprises steps for aborting and safing the system in the event of an anomaly during the Fill System Check Procedure
- The *Static Fire Test Procedure* comprises steps for operating the fill system and conducting a static fire test of the engine.
- The *Static Fire Abort* comprises steps for aborting and safing the system in the event of an anomaly during the Static Fire Test procedure.

Personnel Required

The test operations team consists of ten personnel:

- The **Operations Director [OPS]** directs operations and communicates with all other test personnel
- The **Primary Operator [PRIMARY]** is the main system operator. **PRIMARY** operates all manual valves on the fill and nitrogen pressurization system.
- The **Secondary Operator [SECONDARY]** is the backup for **PRIMARY**, and communicates with **OPS**. If **PRIMARY** becomes incapacitated, **SECONDARY** is responsible for removing them from danger.
- The **Control System Operator [CONTROL]** operates the test control system, including actuating remote valves and engine ignition.
- The **Data Acquisition System Operator [DAQ]** monitors and operates the data acquisition system.
- **Perimeter Guard 1 [P1]**, **Perimeter Guard 2 [P2]**, **Perimeter Guard 3 [P3]** and **Perimeter Guard 4 [P4]** ensure no unauthorized personnel enter the testing area during test operations.
- The **Emergency Services Caller [EMERGENCY]** is responsible for calling 911 in the event of an emergency.

General Notes

- In the event of loss of radio contact, personnel should contact **OPS** by phone at 249-385-6237.
- For **PRIMARY** and **SECONDARY**, if contact with **OPS** cannot be re-established in 1 minute, close all cylinders and return to mission control.
- In the event of an emergency requiring 911 to be called, first responders should be directed to **OPS** before entering the site for a briefing on hazards.
- Unless otherwise specified, any event failing to occur is grounds for abort.

Sign-Off

To be completed by all test personnel after reading and familiarization with procedures

- **Operations Director [OPS]** _____
- **Primary Operator [PRIMARY]** _____
- **Secondary Operator [SECONDARY]** _____
- **Control System Operator [CONTROL]** _____
- **Data Acquisition System Operator [DAQ]** _____
- **Perimeter Guard 1 [P1]** _____
- **Perimeter Guard 2 [P2]** _____
- **Perimeter Guard 3 [P3]** _____
- **Perimeter Guard 4 [P4]** _____
- **Emergency Services Caller [EMERGENCY]** _____

Prior to Fill System Leak Check Start

- ☐ Ensure the following items are complete before beginning the test procedure
  - ☐ Injector assembly.
  - ☐ Concentric tank assembly, including pre-filling fuel tank with 3.75kg of ethanol.
  - ☐ Plumbing setup per Eridium LSF2 Leak Check P&ID.
  - ☐ Data acquisition system setup.
  - ☐ Valve safe state checks.
  - ☐ RLCS power on, including:
    - ☐ Both lipos installed.
    - ☐ Actuator and main switches flipped.
    - ☐ GSPD on and set to AC mode.
  - ☐ Perimeter sweep.
  - ☐ Primary and secondary vent line checks.
  - ☐ Correct plumbing is connected to cylinders and caps are in place.
  - ☐ Generator filled with fuel and running.
  - ☐ System control key inserted in towerside box.
  - ☐ Set all regulators to the minimum position.
  - ☐ Ropes installed on all manual ball valves and tested.
  - ☐ Place an appropriately sized container below Manual Fuel Dump Valve (FMV-101) to collect ethanol in the event dumping is needed.
  - ☐ Low pressure leak check using nitrogen at approx. 100psi and soapy water.
  - ☐ Ensure compressor is turned on and fully pressurized.
  - ☐ Ensure a spray bottle of soapy water is at the test stand.
  - ☐ Verify all valve ropes are fully unrolled.
  - ☐ Ensure all personnel as defined above are available and have completed sign-off.
  - ☐ Ensure a blank copy of the procedure is available or a unique symbol will be used to cross off steps.
  - ☐ Verify the lockout key is inserted into the towerside box.
  - ☐ Ensure the following personnel have walkie-talkies and that communication is functional:
    - ☐ **OPS**
    - ☐ **SECONDARY**
    - ☐ **P1**
    - ☐ **P2**
    - ☐ **P3**
    - ☐ **P4**
  - ☐ Ensure **P1**, **P2**, **P3**, and **P4** are wearing hi-vis vests.
  - ☐ Ensure **PRIMARY** and **SECONDARY** are wearing face shields and have no exposed skin.
  - ☐ Ensure **PRIMARY** is wearing thermal gloves.
  - ☐ Remove propane cylinder from shipping container, if present.

Fill System Leak Check Procedure

- ☐ **PRIMARY**: Open Manual Fuel Dump Valve (FMV-101).
- ☐ **PRIMARY**: Fill ethanol through Manual Fuel Dump Valve (FMV-101) until no more can be added. Be careful not to spill any ethanol.
- ☐ **PRIMARY**: Close Manual Fuel Dump Valve (FMV-101).
- ☐ **PRIMARY**: Open Manual Fuel Tank Isolation Valve (FMV-102).
- ☐ **DAQ**: Zero load cell.
- ☐ **DAQ**: Begin Data logging.
- ☐ **DAQ**: Confirm the zero points of all sensors have been recorded.
- ☐ **PRIMARY**: Confirm the following valves are initially closed:
  - ☐ Remote Oxidizer Fill Valve (OV-101).
  - ☐ Manual Oxidizer Fill Bypass Valve (OMV-101).
  - ☐ Remote Oxidizer Drain Valve (OV-102).
  - ☐ Manual Oxidizer Drain Valve (OMV-102).
  - ☐ Manual Oxidizer Tank Vent Valve (OMV-201).
  - ☐ Nitrogen Purge Valve (NV-101).
  - ☐ Manual Fuel Dump Valve (FMV-101).
  - ☐ Manual Pneumatics Drain Valve (NMV-101).
- ☐ **PRIMARY**: Confirm the following valves are initially open:
  - ☐ Manual Fuel Tank Isolation Valve (FMV-102).
  - ☐ Oxidizer Tank Isolation Valve (OV-201).
- ☐ **CONTROL**: Confirm the commanded state for the following valves is 'closed':
  - ☐ Injector Valve (IJV-101).
- ☐ **CONTROL**: Confirm the commanded state for the following valves is 'open':
  - ☐ Oxidizer Tank Vent Valve (OV-202).
- ☐ **DAQ**: If state feedback is available, confirm the following valves are initially closed:
  - ☐ Fuel Injector Valve (FV-101).
  - ☐ Oxidizer Injector Valve (OV-301).
- ☐ **SECONDARY**: Confirm there are no personnel in the test exclusion zone except for **PRIMARY** and **SECONDARY**.
- ☐ **P1**, **P2**, **P3**, **P4**: Close the perimeter and do not allow any individuals to enter the test exclusion zone.
- ☐ **PRIMARY**: Remove covers from plumbing in the following locations:
  - ☐ Oxidizer Fill Panel PRV (OPRV-101).
  - ☐ Oxidizer fill panel drain hose.
  - ☐ Oxidizer run tank PRV (OPRV-201).
  - ☐ Nitrogen Fill panel PRV (NPRV-101).
  - ☐ Oxidizer Tank Vent Valve (OV-202).

35 ☐ Manual Fuel Dump Valve (FMV-101).

36 ☐ Manual Pneumatics Drain Valve (NMV-101)

37 ☐ **PRIMARY**: Slowly adjust compressor regulator NREG-201 until a pressure of approximately 145psi is read by the downstream gauge (NPG-202).

38 ☐ **PRIMARY**: Communicate final pneumatics pressure as read by NPG-201.

39 ☐ **DAQ**: Confirm pressure reading on NPT-101 is in agreement.

40 ☐ **PRIMARY**: Slowly open nitrogen supply cylinder NSC-101.

41 ☐ **PRIMARY**: Comunicate cylinder pressure as read by NPG-101.

42 ☐ **PRIMARY**: Slowly adjust pneumatics regulator NREG-101 until a pressure of 750psi is read on the downstream gauge (NPG-102).

43 ☐ **DAQ**: Confirm pressure reading on OPT-101 is in agreement.

44 ☐ **PRIMARY** and **SECONDARY**: Spray soapy water on all fittings.

45 ☐ **SECONDARY**: Remove lockout key from towerside box.

46 ☐ **PRIMARY** and **SECONDARY**: Retreat to safe distance.

47 ☐ **CONTROL**: Insert system control key into clientside box and engage control.

48 ☐ **CONTROL**: Close Oxidizer Tank Vent Valve (OV-202).

- If OV-202 fails to close, close Oxidizer Tank Isolation Valve (OV-201).
- If OV-201 fails to close, proceed to the Fill System Leak Check Abort Procedure.

49 ☐ **CONTROL**: Open Remote Oxidizer Fill Valve (OV-101).

- If OV-101 fails to open proceed to the Fill System Leak Check Abort Procedure.

50 ☐ **DAQ**: Communicate fuel and oxidizer tank pressure continuously as read by OPT-201 and FPT-101.

51 ☐ **OPS**: Proceed when oxidizer tank pressure reaches pressure of 750psi.

52 ☐ **OPS**: Record pressure difference between OPT-201 and FPT-101.

53 ☐ **CONTROL**: Close Remote Oxidizer Fill Valve (OV-101).

54 ☐ **DAQ**: Monitor oxidizer and fuel tank pressures.

- If leaks are observed, close Oxidizer Tank Isolation Valve (OV-201)and determine if the leak rate slows.
- If leak rate does not slow after 5 minutes, proceed.

55 ☐ **CONTROL**: Open Oxidizer Tank Vent Valve (OV-202).

56 ☐ **CONTROL**: Open Oxidizer Tank Isolation Valve (OV-201) if closed.

57 ☐ **CONTROL**: Open Remote Oxidizer Drain Valve (OV-102).

58 ☐ **OPS**: Proceed when oxidizer tank is fully vented.

59 ☐ **CONTROL**: Disengage control and remove system control key from clientside box.

60 ☐ **PRIMARY** and **SECONDARY**: Return to the test stand.

61 ☐ **SECONDARY**: Insert lockout key into towerside box.

- ☐ **PRIMARY**: Inspect all plumbing for evidence of bubbles. Report findings to **OPS**.
- ☐ **OPS**: If leaks were discovered that can be easily fixed, have **PRIMARY** tighten the fittings and return to Line 39.
- ☐ **PRIMARY**: Close Nitrogen supply cylinder NSC-101.
- ☐ **PRIMARY**: Set compressor regulator NREG-201 to minimum position.
- ☐ **PRIMARY**: Slowly open Manual Pneumatics Drain Valve (NMV-101).
- ☐ **SECONDARY**: Remove lockout key from towerside box.
- ☐ **PRIMARY** and **SECONDARY**: Retreat to safe distance.
- ☐ **CONTROL**: Insert system control key into clientside box and engage control.
- ☐ **CONTROL**: Open Remote Oxidizer Fill Valve (OV-101).
- ☐ **DAQ**: Confirm all sensors are reading their determined zero points.
- ☐ **CONTROL**: Disengage control and remove system control key from the clientside box.
- ☐ **P1**, **P2**, **P3**, and **P4**: Open the perimeter.
- ☐ **DAQ**: Stop DAQ logging and save data files.
- ☐ **PRIMARY** and **SECONDARY**: Approach the test stand.
- ☐ **PRIMARY**: Replace covers on plumbing in the following 7 locations:
    - ☐ Oxidizer fill panel PRV (OPRV-101).
    - ☐ Oxidizer fill panel drain hose.
    - ☐ Oxidizer run tank PRV (OPRV-201).
    - ☐ Nitrogen Fill panel PRV (NPRV-101)
    - ☐ Oxidizer Tank Vent Valve (OV-202).
    - ☐ Manual Fuel Dump Valve (FMV-101).
    - ☐ Manual Pneumatics Drain Valve (NMV-101)
- ☐ **OPS**: If leaks were observed an not fixed, correct leaks and repeat procedure as needed. Once no leaks are present, proceed with final setup for Static Fire Test.

Fill System Leak Check Abort Procedure

1. ☐ **CONTROL**: Open Oxidizer Tank Vent Valve (OV-202) if not already open.
   - If RLCS is not operational, have **PRIMARY** open Manual Oxidizer Drain Valve (OMV-102) using the attached rope.
2. ☐ **CONTROL**: Open Oxidizer Tank Isolation Valve (OV-201) if not already open.
3. ☐ **CONTROL**: Close Remote Oxidizer Fill Valve (OV-101) if not already closed.
4. ☐ **CONTROL**: Open Remote Oxidizer Drain Valve (OV-102).
5. ☐ **CONTROL**: Remove system control key and disengage control.
6. ☐ **OPS**: Proceed when oxidizer tank is fully vented.
7. ☐ **PRIMARY** and **SECONDARY**: Approach the test stand.
8. ☐ **SECONDARY**: Insert lockout key into towerside box.
9. ☐ **PRIMARY**: Close nitrogen supply cylinder NSC-101.
10. ☐ **PRIMARY**: Turn off the air compressor.
11. ☐ **PRIMARY**: Set the air compressor regulator to zero.
12. ☐ **SECONDARY**: Remove lockout key from towerside box.
13. ☐ **PRIMARY** and **SECONDARY**: Retreat to a safe distance.
14. ☐ **CONTROL**: Insert system control key and engage control.
15. ☐ **CONTROL**: Open Remote Oxidizer Fill Valve (OV-101).
    - If RLCS is not operational, have **PRIMARY** open Manual Oxidizer Fill Bypass Valve (OMV-101) using the attached rope.
16. ☐ **CONTROL**: Open Nitrogen Purge Valve (NV-101).
    - If RLCS is not operational, proceed. Nitrogen will be drained through Manual Pneumatics Drain Valve (NMV-101).
17. ☐ **CONTROL**: Disengage control and remove the system control key.
18. ☐ **DAQ**: Confirm all sensors are reading their determined zero points.
    - If DAQ is not operational, have **PRIMARY** verify the gauges OPG-101, OPG-201 and NPG-101 are all reading zero.
19. ☐ **PRIMARY**: Open Manual Pneumatics Drain Valve (NMV-101).
20. ☐ **P1**, **P2**, **P3**, and **P4**: Open the perimeter.
21. ☐ **OPS**: Abort and revisit the plumbing and controls setup.

Prior to Static Fire Test Start

- ☐ Ensure the following items are complete before beginning the test procedure
  - ☐ Fill System Leak Check procedure complete.
  - ☐ Combustion chamber assembly.
  - ☐ Plumbing setup per Eridium LSF2 P&ID.
  - ☐ Data acquisition system setup.
  - ☐ Valve safe state checks.
  - ☐ RLCS power on, including:
    - ☐ Both lipos installed.
    - ☐ Actuator and main switches flipped.
    - ☐ GSPD on and set to AC mode.
  - ☐ Perimeter sweep.
  - ☐ Primary and secondary vent line checks.
  - ☐ Correct plumbing is connected to cylinders and caps are in place.
  - ☐ Tank heating jackets installed on oxidizer run tank and set to 60°C.
  - ☐ Cameras in the correct positions and prepared to record.
  - ☐ Generator filled with fuel and running.
  - ☐ System control key inserted in towerside box.
  - ☐ Set all regulators to the minimum position.
  - ☐ Ensure ropes are installed on manual ball valves and tested.
  - ☐ Place an appropriately sized container below Manual Fuel Dump Valve (FMV-101) to collect ethanol in the event dumping is needed.
  - ☐ Ensure fuel tank is filled with 3.75kg of ethanol.
  - ☐ Ensure all personnel as defined above are available and have completed sign-off.
  - ☐ Ensure a blank copy of the procedure is available or a unique symbol will be used to cross off steps.
  - ☐ Verify lockout key is installed in towerside box.
  - ☐ Verify ignition cables are disconnected from the towerside box.
  - ☐ Verify a multimeter is at the test tand.
  - ☐ Verify all valve ropes are fully unrolled.
  - ☐ Ensure the following personnel have walkie-talkies and that communication is functional:
    - ☐ **OPS**
    - ☐ **SECONDARY**
    - ☐ **P1**
    - ☐ **P2**
    - ☐ **P3**
    - ☐ **P4**
  - ☐ Ensure **P1**, **P2**, **P3**, and **P4** are wearing hi-vis vests.
  - ☐ Ensure **PRIMARY** and **SECONDARY** are wearing face shields and have no exposed skin.
  - ☐ Ensure **PRIMARY** is wearing thermal gloves.
  - ☐ Remove propane heater from shipping container, if present.

Static Fire Test Procedure

- ☐ **DAQ**: Begin DAQ data logging.
- ☐ **SECONDARY**: Confirm that the ignition wire is not connected to the towerside control box.
- ☐ **DAQ**: Confirm the zero points of all sensors have been recorded.
- ☐ **PRIMARY**: Confirm the following valves are initially closed:
  - ☐ Remote Oxidizer Fill Valve (OV-101).
  - ☐ Manual Oxidizer Fill Bypass Valve (OMV-101).
  - ☐ Remote Oxidizer Drain Valve (OV-102).
  - ☐ Manual Oxidizer Drain Valve (OMV-102).
  - ☐ Nitrogen Purge Valve (NV-101).
  - ☐ Manual Pneumatics Drain Valve (NMV-101).
  - ☐ Manual Oxidizer Tank Vent Valve (OMV-201).
  - ☐ Manual Fuel Dump Valve (FMV-101).
- ☐ **PRIMARY**: Confirm the following valves are initially open:
  - ☐ Manual Fuel Tank Isolation Valve (FMV-102).
  - ☐ Oxidizer Tank Isolation Valve (OV-201).
- ☐ **CONTROL**: Confirm the commanded state for the following valves is 'closed':
  - ☐ Injector Valve (IJV-101).
- ☐ **CONTROL**: Confirm the commanded state for the following valves is 'open':
  - ☐ Oxidizer Tank Vent Valve (OV-202).
- ☐ **DAQ**: If state feedback is available, confirm the following valves are initially closed:
  - ☐ Fuel Injector Valve (FV-101).
  - ☐ Oxidizer Injector Valve (OV-301).
- ☐ **PRIMARY**: Confirm the resistance across the primary and secondary ignition cables is 2.0 +/- 0.5 ohms.
- ☐ **SECONDARY**: Confirm there are no personnel in the test exclusion zone except for **PRIMARY** and **SECONDARY**.
- ☐ **DAQ**: Confirm all sensors are reading their determined zero points.
- ☐ **P1**, **P2**, **P3**, **P4**: Close the perimeter and do not allow any individuals to enter the test exclusion zone.
- ☐ **PRIMARY**: Confirm all covers are removed from the system. There should be 8 covers total, with covers in the following locations:
  - ☐ Oxidizer fill panel PRV (OPRV-101).
  - ☐ Oxidizer fill panel drain hose.
  - ☐ Oxidizer run tank PRV (OPRV-201).
  - ☐ Nitrogen Fill Panel PRV (NPRV-101).
  - ☐ Oxidizer Tank Vent Valve (OV-202).
  - ☐ Manual Pneumatics Drain Valve (NMV-101).
  - ☐ Manual Fuel Dump Valve (FMV-101).

- ☐ Combustion chamber/nozzle.

☐ **PRIMARY**: Slowly open Nitrogen Supply Cylinder NSC-101.

☐ **PRIMARY**: Communicate cylinder supply pressure as read by NPG-101.

☐ **PRIMARY**: Slowly adjust pneumatics regulator NREG-101 until a pressure of 145psi is read on NPG-102.

☐ **DAQ**: Confirm the pressure reading on NPT-101 is 145psi.

☐ **PRIMARY**: Slowly open the first Nitrous Oxide supply cylinder, OSC-101.

- If leaks are observed proceed to the Static Fire Remote Abort Procedure.

☐ **PRIMARY**: Slowly open the second Nitrous Oxide supply cylinder, OSC-102.

☐ **PRIMARY**: Communicate the oxidizer supply pressure as read by gauge OPG-101.

☐ **DAQ**: Communicate the oxidizer supply pressure as read by OPT-101.

☐ **OPS**: Confirm the two values are in agreement.

☐ **PRIMARY**: Start recording on all manual cameras.

☐ **SECONDARY**: Connect the ignition wire to the towerside box.

☐ **SECONDARY**: Remove lockout key from towerside box.

☐ **PRIMARY** and **SECONDARY**: Retreat to mission control. Bring the multimeter.

☐ **CONTROL**: Insert system control key into clientside box and engage control.

☐ **CONTROL**: Open Remote Oxidizer Drain Valve (OV-102).

☐ **CONTROL**: Open Nitrogen Purge Valve (NV-101) to begin nitrogen purge of plumbing.

☐ **OPS**: Proceed after 1 minute.

☐ **CONTROL**: Close Remote Oxidizer Drain Valve (OV-102).

☐ **CONTROL**: Close Nitrogen Purge Valve (NV-101).

☐ **CONTROL**: Open Remote Oxidizer Fill Valve (OV-101).

- If Remote Oxidizer Fill Valve (OV-101) fails to the Static Fire Remote Abort Procedure.

☐ **DAQ**: Continuously communicate oxidizer tank mass as read by oxidizer tank load cell.

☐ **OPS**: Proceed when tank mass has increased by 13.5 kg (+ 0.5kg, - 0 kg), or a plume is observed from Oxidizer Tank Vent Valve (OV-202).

- If supply and oxidizer tank pressure equilibrate before target conditions are met proceed to the Static Fire Remote Abort Procedure.

☐ **CONTROL**: Close Remote Oxidizer Fill Valve (OV-101).

- If Remote Oxidizer Fill Valve (OV-101) fails to close proceed to the Static Fire Remote Abort Procedure.

☐ **CONTROL**: Close Oxidizer Tank Vent Valve (OV-202).

- If OV-202 fails to close, close Oxidizer Tank Isolation Valve (OV-201).
- If Oxidizer Tank Isolation Valve (OV-201) fails to close, proceed to the Static Fire Remote Abort Procedure.

- ☐ **CONTROL**: Engage tank heating.
- ☐ **DAQ**: Communicate oxidizer tank pressure as read by OPT-201 in increments of 50psi.
- ☐ **DAQ**: Monitor all sensors and report any deviations or anomalies to **OPS** immediately.
- ☐ **OPS**: If oxidizer tank mass is decreasing or leaking from Oxidizer Tank Vent Valve (OV-202) is suspected, close Oxidizer Tank Isolation Valve (OV-201).
- ☐ **OPS**: If oxidizer tank mass drops below 12 kg proceed to the Static Fire abort procedure,
- ☐ **OPS**: Proceed when the oxidizer tank pressure has reached 700psi.
- ☐ **OPS**: Ensure all spectators are wearing safety glasses and have earplugs in.
- ☐ **OPS**: Verify all remote cameras are recording.
- ☐ **OPS**: Instruct drone operator to launch.
- ☐ **OPS**: Proceed when the oxidizer tank pressure has reached firing pressure of 750psi.
- ☐ **CONTROL**: Disengage tank heating.
- ☐ **PAUSE POINT**
- ☐ **OPS**: Verify all test commit criteria are met.
- ☐ **BEGIN RADIO SILENCE** - Note that in the event of an emergency or anomaly requiring immediate attention radio silence will be broken.
- ☐ **CONTROL**: Conduct the engine startup procedure.
  - ☐ Arm primary ignition.
  - ☐ Hold down the fire button and continuously communicate ignition current.
  - ☐ Proceed when either primary or secondary ignition current reads 0 amps, or when smoke is observed.
    - In the event of a failed ignition (criteria are not met within 1 minute) immediately release the ignition button and proceed to the Static Fire Remote Abort Procedure
  - ☐ Conduct the static fire test by opening Injector Valve (IJV-101).
    - In the event Injector Valve (IJV-101) does not open proceed to the Static Fire Remote Abort Procedure.
  - ☐ **ALL**: Observe the plume.
- ☐ **OPS**: Proceed when the static fire has concluded.
- ☐ **DAQ**: Verify DAQ systems are functional and all sensors are transmitting data.
- ☐ **OPS**: Confirm with remote camera operators that there is no visible fire on or around the test stand.
  - In the event fire is observed:
    - ☐ **OPS**: If the fire is major or outside of the fill area, **call 911 immediately**.
    - ☐ **OPS**: If fire is localized to the engine, proceed with nitrogen purge. Repeat as needed until the fire is extinguished.
    - ☐ If a small fire is present in the fill area that cannot be extinguishers by a nitrogen purge, send **PRIMARY** and **SECONDARY** to use fire extinguishers.
- ☐ **CONTROL**: Conduct a nitrogen purge by opening Nitrogen Purge Valve (NV-101).
- ☐ **CONTROL**: Close Nitrogen Purge Valve (NV-101).

- ☐ **CONTROL**: Close Injector Valve (IJV-101).
- ☐ **END RADIO SILENCE**
- ☐ **CONTROL**: Open Oxidizer Tank Vent Valve (OV-202).
- ☐ **CONTROL**: Open Oxidizer Tank Isolation Valve (OV-201) if closed.
- ☐ **CONTROL**: Disengage control and remove system control key.
- ☐ **DAQ**: Confirm CC temperature as read by CCTC-101, -102, -103, and/or -104 is below 100°C, unless suspected faulty.
- ☐ **PRIMARY** and **SECONDARY**: Approach the test stand.
- ☐ **SECONDARY**: Insert lockout key into towerside box.
- ☐ **PRIMARY**: Close Nitrous Oxide supply cylinder OSC-101.
- ☐ **PRIMARY**: Close Nitrous Oxide supply cylinder OSC-102.
- ☐ **PRIMARY**: Close Nitrogen supply cylinder NSC-101.
- ☐ **SECONDARY**: Remove lockout key from towerside box.
- ☐ **PRIMARY** and **SECONDARY**: Retreat to mission control.
- ☐ **CONTROL**: Insert system control key into clientside box and engage control.
- ☐ **CONTROL**: Open Remote Oxidizer Drain Valve (OV-102).
- ☐ **CONTROL**: Open Remote Oxidizer Fill Valve (OV-101).
- ☐ **CONTROL**: Open Nitrogen Purge Valve (NV-101).
- ☐ **DAQ**: Confirm all sensors are reading their determined zero points.
- ☐ **CONTROL**: Disengage control and remove system control key from the clientside box.
- ☐ **DAQ**: Stop DAQ logging and save data files.
- ☐ **PRIMARY** and **SECONDARY**: Approach the test stand.
- ☐ **SECONDARY**: Insert lockout key into towerside box.
- ☐ **PRIMARY**: Open Manual Pneumatics Drain Valve (NMV-101).
- ☐ **P1**, **P2**, **P3**, and **P4**: Open the perimeter.
- ☐ **PRIMARY**: Replace plumbing covers in the following 7 locations:
  - ☐ Oxidizer fill panel PRV (OPRV-101).
  - ☐ Oxidizer fill panel drain hose.
  - ☐ Oxidizer run tank PRV (OPRV-201).
  - ☐ Nitrogen Panel PRV (NPRV-101)
  - ☐ Oxidizer Tank Vent Valve (OV-202).
  - ☐ Manual Fuel Dump Valve (FMV-101).
  - ☐ Manual Pneumatics Drain Valve (NMV-101).
- ☐ **DAQ**: Continue to monitor CC temperature. Inform **OPS** immediately if temperature exceeds 100°C.
- ☐ **OPS**: Proceed with teardown.

Static Fire Remote Abort Procedure

1. ☐ **CONTROL**: Release the ignition button if pressed.
2. ☐ **DAQ**: Verify DAQ is operational and receiving data.
    - If DAQ or RLCS is not operational, proceed to the Static Fire Manual Abort Procedure.
3. ☐ **CONTROL**: Disengage tank heating.
4. ☐ **CONTROL**: Open Oxidizer Tank Vent Valve (OV-202) if not already open.
5. ☐ **CONTROL**: Open Oxidizer Tank Isolation Valve (OV-201) if not already open.
6. ☐ **CONTROL**: Close Remote Oxidizer Fill Valve (OV-101) if not already closed.
7. ☐ **CONTROL**: Close Nitrogen Purge Valve (NV-101) if not already closed.
8. ☐ **CONTROL**: Open Remote Oxidizer Drain Valve (OV-102).
9. ☐ **OPS**: Proceed when the oxidizer tank is fully vented.
10. ☐ **CONTROL**: Close Injector Valve (IJV-101) if not already closed.
11. ☐ **CONTROL**: Disengage control and remove the system control key.
12. ☐ **PRIMARY** and **SECONDARY**: Approach the test stand.
13. ☐ **SECONDARY**: Insert lockout key into towerside box.
14. ☐ **SECONDARY**: Disconnect the ignition wires from the towerside box.
15. ☐ **PRIMARY**: Close Nitrous Oxide supply cylinder OSC-101.
16. ☐ **PRIMARY**: Close Nitrous Oxide supply cylinder OSC-102.
17. ☐ **PRIMARY**: Close Nitrogen supply cylinder NSC-101.
18. ☐ **SECONDARY**: Remove lockout key from towerside box.
19. ☐ **PRIMARY** and **SECONDARY**: Retreat to a safe distance.
20. ☐ **CONTROL**: Insert the system control key and engage control.
21. ☐ **CONTROL**: Open Remote Oxidizer Fill Valve (OV-101).
22. ☐ **CONTROL**: Open Nitrogen Purge Valve (NV-101).
23. ☐ **DAQ**: Verify all sensors are reading their determined zero point.
24. ☐ **CONTROL**: Disengage control and remove system control key.
25. ☐ **PRIMARY** and **SECONDARY**: Approach the test stand.
26. ☐ **PRIMARY**: Open Manual Pneumatics Drain Valve (NMV-101).
    - If ethanol draining is required, have **PRIMARY** place a bucket underneath the outlet hose from Manual Fuel Dump Valve (FMV-101), then open Manual Fuel Dump Valve (FMV-101).
27. ☐ **P1**, **P2**, **P3**, and **P4**: Open the perimeter.
28. ☐ **OPS**: Abort and revisit plumbing and controls setup.

Static Fire Manual Abort Procedure

1. ☐ **CONTROL** and **DAQ**: Attempt to troubleshoot the issue.
   - If RLCS and DAQ become operational, have **PRIMARY** and **SECONDARY** retreat and return to the Static Fire Remote Abort Procedure.
   - If a connection issue is suspected send a DAQ technician with a laptop and walkie talkie to the manual valve pull ropes and attempt to connect to the local wifi. If successful, proceed with a remote abort.
2. ☐ **PRIMARY** and **SECONDARY**: Approach the manual valve ropes and listen for venting.
   - If venting is heard, report approximate source to **OPS**, and do not proceed until cleared by **OPS**.
3. ☐ **PRIMARY**: Open Manual Oxidizer Drain Valve (OMV-102) using the attached rope.
   - If no venting is heard or Manual Oxidizer Drain Valve (OMV-102) fails to open, open Manual Oxidizer Tank Vent Valve (OMV-201).
4. ☐ **OPS**: Proceed when **PRIMARY** and **SECONDARY** report no venting is heard.
5. ☐ **SECONDARY**: Insert lockout key into towerside box.
6. ☐ **SECONDARY**: Disconnect the ignition wires from the towerside box.
7. ☐ **PRIMARY**: Close Nitrous Oxide supply cylinder OSC-101.
8. ☐ **PRIMARY**: Close Nitrous Oxide supply cylinder OSC-102.
9. ☐ **PRIMARY**: Close Nitrogen supply cylinder NSC-101.
10. ☐ **PRIMARY** and **SECONDARY**: Retreat to the manual valve ropes.
11. ☐ **PRIMARY**: Open Manual Oxidizer Fill Bypass Valve (OMV-101).
12. ☐ **OPS**: Proceed when **PRIMARY** and **SECONDARY** report no venting is heard.
13. ☐ **PRIMARY**: Open Manual Pneumatics Drain Valve (NMV-101).
14. ☐ **PRIMARY**: Verify the following gauges read zero pressure:
    - Oxidizer fill panel gauge OPG-101.
    - Nitrogen fill panel gauge NPG-101.
    - Vent panel gauge OPG-201.
    - If ethanol draining is required, have **PRIMARY** place a bucket underneath the outlet hose from Manual Fuel Dump Valve (FMV-101), then open Manual Fuel Dump Valve (FMV-101).
15. ☐ **P1**, **P2**, **P3**, and **P4**: Open the perimeter.
16. ☐ **OPS**: Abort and revisit plumbing and controls setup.

Static Fire Emergency Abort Procedure

1. ☐ **CONTROL**: Release the ignition button if pressed.
2. ☐ **CONTROL**: Open Remote Oxidizer Drain Valve (OV-102).
   - If DAQ or RLCS is not operational and it is safe to approach, have **PRIMARY** open Manual Oxidizer Drain Valve (OMV-102) using the attached rope.
   - If no venting occurs when Manual Oxidizer Drain Valve (OMV-102) is opened or Manual Oxidizer Drain Valve (OMV-102) fails to open, have **PRIMARY** open Manual Oxidizer Tank Vent Valve (OMV-201) using the attached rope.
   - If is it not safe to approach the test site, call 911. 911 should be directed to **OPS** before approaching the site.
3. ☐ **CONTROL**: Close Remote Oxidizer Fill Valve (OV-101) if not already closed.
4. ☐ **CONTROL**: Open Oxidizer Tank Vent Valve (OV-202) if not already open.
5. ☐ **CONTROL**: Open Oxidizer Tank Isolation Valve (OV-201) if not already open.
6. ☐ **CONTROL**: Disengage tank heating.
   - If DAQ or RLCS is not operational, have **PRIMARY** turn off the generator.
7. ☐ **OPS**: Proceed when the oxidizer run tank is fully vented.
8. ☐ **CONTROL**: Close Injector Valve (IJV-101) if not already closed.
9. ☐ **CONTROL**: Remove system control key and disengage control.
10. ☐ **PRIMARY** and **SECONDARY**: Approach the test stand.
11. ☐ **SECONDARY**: Insert the lockout key into the towerside box.
12. ☐ **SECONDARY**: Disconnect the ignition wires from the towerside box.
13. ☐ **PRIMARY**: Close Nitrous Oxide supply cylinder OSC-101.
14. ☐ **PRIMARY**: Close Nitrous Oxide supply cylinder OSC-102.
15. ☐ **PRIMARY**: Close Nitrogen supply cylinder NSC-101.
16. ☐ **SECONDARY**: Remove system control key from the towerside box.
17. ☐ **PRIMARY** and **SECONDARY**: Retreat to a safe distance.
18. ☐ **CONTROL**: Insert the system control key and engage control.
19. ☐ **CONTROL**: Open Remote Oxidizer Fill Valve (OV-101).
20. ☐ **CONTROL**: Open Nitrogen Purge Valve (NV-101).
21. ☐ **DAQ**: Verify all sensors are reading their determined zero point.
    - If DAQ is not operational,**PRIMARY** and **SECONDARY**. are not to approach the test stand if venting is heard.
22. ☐ **CONTROL**: Disengage control and remove system control key.
23. ☐ **PRIMARY** and **SECONDARY**: Approach the test stand.
24. ☐ **PRIMARY**: Open Manual Pneumatics Drain Valve (NMV-101).

- If ethanol draining is required, have **PRIMARY** place a bucket underneath the outlet hose from Manual Fuel Dump Valve (FMV-101), then open Manual Fuel Dump Valve (FMV-101).

25 ☐ **P1**, **P2**, **P3**, and **P4**: Open the perimeter.

26 ☐ **OPS**: Abort and revisit plumbing and controls setup.

Variable Recording

PT1 (Ox Fill) zero point		
PT2 (Ox Tank) zero point		
PT3 (Nitrogen Fill) zero point		
PT4 (Fuel Tank) zero point		
PT5 (Pneumatics) zero point		
PT6 (Ox injector) zero point		
PT7 (Fuel injector) zero point		
PT8 (CC) zero point		
PT9 (CC) zero point		
Ox tank load cell zero point		
Fuel tank load cell zero point		
CC load cell zero point		
Pneumatic pressure		